

Activity 3

Aim: Calibration of pipette by burette.

Apparatus: Burette (25 mL), pipette (10 mL), beaker (100 mL), etc.

Chemicals: Distilled water

Procedure:

1. Take a 25 mL burette and fixed it on burette stand.
2. Fill the burette with distilled water upto 10 mL mark with lower meniscus.
3. Now pipette out 10 mL of distilled water and add it to the burette.
4. Observe the level of water in the burette, if it is zero then that pipette is calibrated.
5. If the level of the water in the burette is not zero or more than zero, then take another pipette for calibration.
6. Continue this process till pipette is calibrated.

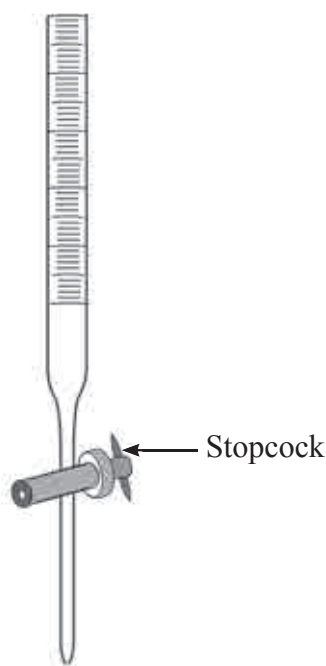


Fig. Activity 3.1 (a) Burette
(25 mL capacity)

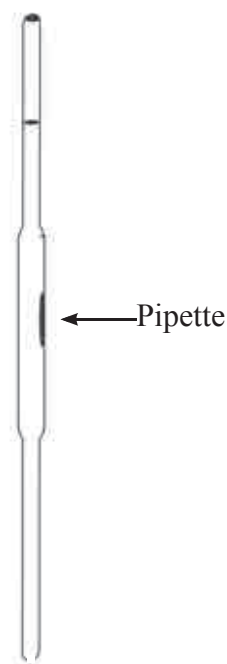


Fig. Activity 3.1 (b) Pipette
(10 mL capacity)

Conclusion: The pipette is calibrated.

Remark and sign of teacher:

MCQ

Select [✓] or underline the most appropriate answer from given alternatives of each sub question.

1. The minimum volume of solution is measured by burette is
 - a. 0.01 mL
 - ☒ b. 0.1 mL
 - c. 1.0 mL
 - d. 10.0 mL
2. The capacity of pipette used by you in this activity is
 - ☒ a. 0.01 L
 - b. 0.1 L
 - c. 1 L
 - d. 10 L
3. The Least count of the burette is
 - a. 1 mL
 - ☒ b. 0.1 mL
 - c. 0.01 mL
 - d. 0.001 mL
4. A student added 10 mL distilled water in burette already having 5 mL of distilled water by calibrated pipette. What is correct burette reading in burette?
 - a. 5 mL
 - ☒ c. 15 mL
 - b. 10 mL
 - d. 25 mL

Short answer questions

1. Arrange the following step in proper sequence
 - a. Burette fix on a burette stand
 - b. Observed the level of water in burette
 - c. Take 25 mL burette
 - d. Take 10 mL of distilled water by pipette
 - e. Add 10 mL water from pipette to the burette
 - f. Fill the burette upto 10 mL with water

Ans.

c. Take 25 mL burette.

a. Fix the burette on a burette stand.

f. Fill the burette up to 10 mL with water.

d. Take 10 mL of distilled water by pipette.

e. Add 10 mL water from pipette to the burette.

b. Observe the level of water in burette.

Remark and sign of teacher: